



Quant Week 5

3/13/25



Agenda



1. Icebreaker Question
2. Brainteasers
3. Zetamac?
4. VWAP & TWAP
5. Project



Ice Breaker

Favorite place to eat in Ann Arbor





Weekly Sequence

2, 3, 6, $\frac{1}{2}$, 3, $\frac{1}{6}$





Brain Teaser - Card Game



You are playing a simple card game in which you and the dealer both draw a card.

You win if your card is higher than theirs

You lose if your card is lower than or equal to theirs

What is the probability that you win?

Brain Teaser - Card Game - Solution



What is the probability that you win?

Each card has a $1/13$ chance of being picked

$$\begin{aligned} 1/13 * (0+4/51+8/51+ \dots +48/51) &= 4/(13*51)*(0+1+ \dots + 13) = 4(13*51)*12*13/2 \\ &= 8/17 \end{aligned}$$

Simpler way:

E_1 = your card is higher

E_2 = your card is the same

E_3 = your card is lower

$$P(E_1) = P(E_3)$$

$$P(E_1) + P(E_2) + P(E_3) = 1$$

Find $P(E_2)$ and then you can calculate it

$$P(E_2) = 3/51 = 1/17$$

$$2*P(E_1) + 1/17 = 1$$

$$P(E_1) = 8/17$$

Schedule-Based Single Order Algorithms





Let's say you are a trader and your firm wants to buy (or sell) a large volume of a single stock.

A large volume in this context is one that will change the price

Order Book Example, Market Order of 1000



Bid (Volume)	Price	Ask (Volume)
	100.05	350
	100.04	450
	100.03	200
	100.02	150
	100.01	
120	100.00	
180	99.99	

Order Book Example, Market Order of 1000



Bid (Volume)	Price	Ask (Volume)
	100.05	350 (150 SOLD)
	100.04	450 (SOLD)
	100.03	200 (SOLD)
	100.02	150 (SOLD)
	100.01	50 (SOLD)
120	100.00	
180	99.99	

Order Book Example



Bid (Volume)	Price	Ask (Volume)
	100.05	200
	100.04	
	100.03	
	100.02	
	100.01	
120	100.00	
180	99.99	



As you can see, the best ask when the market order was placed was 10.01.

Some orders went through at this price, but not all of them.

Some went through as high as 10.05.



Order Book Example

Bid (Volume)	Price	Ask (Volume)
	100.01	
	120.00	200
		
	100.10	50
	100.09	50
	100.08	50
	100.07	60
	100.06	40
	100.05	50
	100.04	200
	100.03	100
	100.02	100
	100.01	100
120	100.00	
180	99.99	



Order Book Example

For a larger example, if there is not much demand at all, transacting may lead to much higher prices.

Here some orders are placed at \$120 because the market is “thin”

thin means relatively few buyers/sellers

Bid (Volume)	Price	Ask (Volume)
	120.01	
	120.00	200 (SOLD)
		
	100.10	50 (SOLD)
	100.09	50 (SOLD)
	100.08	50 (SOLD)
	100.07	60 (SOLD)
	100.06	40 (SOLD)
	100.05	50 (SOLD)
	100.04	200 (SOLD)
	100.03	100 (SOLD)
	100.02	100 (SOLD)
	100.01	100 (SOLD)
120	100.00	
180	99.99	

Order Book Example, Limit Order of 1000



Bid (Volume)	Price	Ask (Volume)
	100.05	350
	100.04	450
	100.03	200
	100.02	150
	100.01	
120	100.00	
180	99.99	

Order Book Example, Limit Order of 1000



Bid (Volume)	Price	Ask (Volume)
	100.05	350
	100.04	450
	100.03	200
	100.02	150
	100.01	50
1,120 (1,120 Yours)	100.00	
180	99.99	



People will be able to see that you placed this order, and others will try to “front run” your order

Others know this order is likely to make prices go up so others will try to buy

They see your order exists and has not transacted yet, so they will try to beat you to it by buying one cent above you

This makes YOU lose out on profits!



How to avoid these things

- Use a combination of both market and limit orders
- Sell over time

TWAP - Time-Weighted Average Price

- Buy/Sell a handful of the stock in intervals
- This way others won't see your large limit orders, and market orders will (hopefully) get better prices



TWAP & VWAP

TWAP - Time-Weighted Average Price

VWAP - Volume-Weighted Average Price

- Match the intervals to corresponding trading volume

Comparison Indicators Technicals Corporate Events

Line



Share Settings

NVDA

P: 913.92 O: 915.22 H: 915.45 L: 913.89 C: 913.92 VOL: 96.9K

Plots

vol undr

96,920



Project Discussion/Work Time

The background is a solid orange color. On the right side, there is a series of parallel diagonal lines running from the top right towards the bottom left. These lines are composed of alternating white and dark blue segments, creating a striped effect. Overlaid on these lines is a grid of squares, some of which are white and others are dark blue, creating a complex geometric pattern.